ENGINEERING & ACADEMIC PROJECT REPORT

RETAIL INVENTORY OPTIMIZATION SYSTEM

A dense, high-frequency multi-facility inventory operations platform engineered with Nothing OS Swiss Industrial Design, Angular 17+ Signals, and Spring Boot 3.3.5 REST Architecture.

REPOSITORY URL

https://github.com/atharav9927/STOCKSMART

LIVE PRODUCTION URL

https://atharav9927.github.io/STOCKSMART/

TECHNOLOGY FOCUS

Angular 17 • Spring Boot 3 • Java 21 • TypeScript

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1. Executive Summary

Modern multi-echelon retail supply chains frequently suffer from fragmented inventory visibility, high wedge latency during barcode capture, cognitive fatigue from bloated enterprise dashboards, and stockout reconciliation delays. **StockSmart** was engineered to solve these challenges through an ultra-dense, low-latency operational control plane.

Designed using the **Nothing OS Swiss Modernist design philosophy** (monochromatic OLED canvas, `Doto` dot-matrix numbers, `Space Grotesk` / `Space Mono` type hierarchy, and zero superficial drop shadows), the system reserves color strictly for actionable alerts (signal red for stockouts and green for healthy velocity).

> **Core Architectural Innovation:** StockSmart operates simultaneously in two production modes:
> - **Autonomous Client-Side SPA:** 100% functional on GitHub Pages with an in-browser reactive signals data engine and persistent local session state (all 42 routes work without a remote server).
> - **Enterprise Cloud Mode:** A robust Spring Boot 3.3.5 REST API with PostgreSQL persistence, stateless JWT security, and Docker/Render container deployment blueprints.

2. Technology Stack & Language Breakdown

2.1. Programming Languages

LANGUAGE	VERSION / SPEC	SYSTEM SCOPE
TypeScript	v5.4.x	Angular 17 standalone components, reactive signals, models, routing guards, interceptors
Java	OpenJDK 21 / 25	Spring Boot 3.3.5 REST API, Spring Security filter chains, JPA entities, JWT encryption
HTML5 / CSS3 / SCSS	W3C Standards	Nothing OS design tokens, OLED theme, responsive grid layouts, thermal label print styles
SQL	ANSI / H2 / PG	Relational schema modeling, multi-facility inventory indexing, immutable ledger queries

2.2. Frameworks & Core Libraries

- **Frontend Framework:** `Angular 17.3` (Standalone Architecture, zero `NgModule` overhead, 65 tree-shaken lazy-loaded route chunks).
- **State Management:** Native `Angular Signals` (`signal()`, `computed()`, `effect()`) delivering instantaneous DOM updates without zone-pollution.

- **Barcode Scanning Engine:** `@zxing/library` & `@zxing/browser` providing real-time WebRTC camera video barcode decoding with an optical HUD targeting reticle.
- **Acoustic Feedback:** Native `Web Audio Oscillator API` generating 880 Hz positive sine beeps and 220 Hz sawtooth alerts.
- **Backend Framework:** `Spring Boot 3.3.5` with Spring Web MVC, Spring Data JPA, and Spring Actuator.
- **Security:** `Spring Security 6` with stateless JSON Web Tokens (`jjwt 0.12.5`) and BCrypt password encryption (strength 12).
- **Databases:** Dual-profile architecture using in-memory `H2 Database` for local development and managed `PostgreSQL 16` on Render.

3. Visual Telemetry & Live Screen Captures

The system is deployed and fully operational at `https://atharav9927.github.io/STOCKSMART/dashboard`. The following screen captures illustrate the live application state:

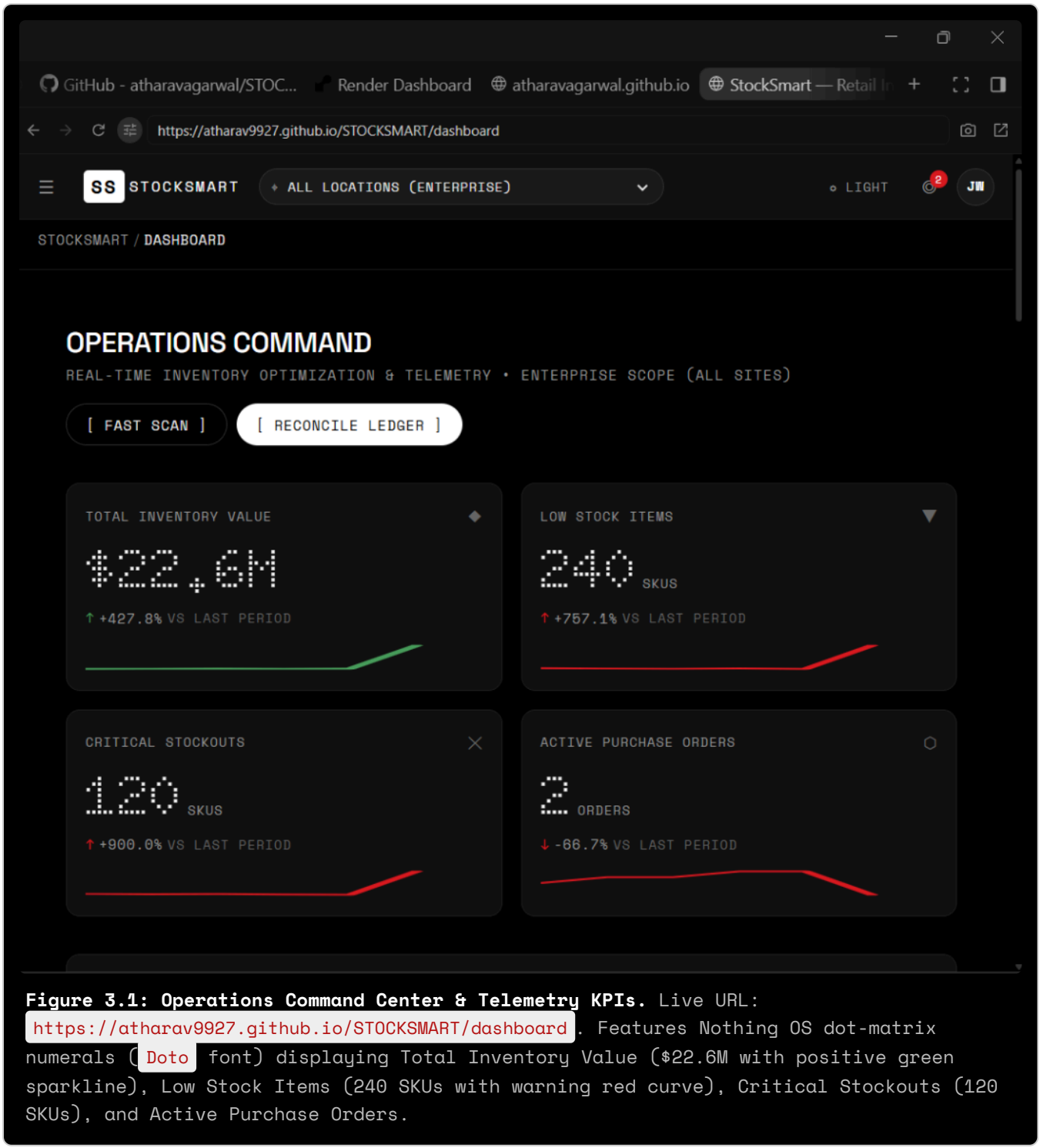
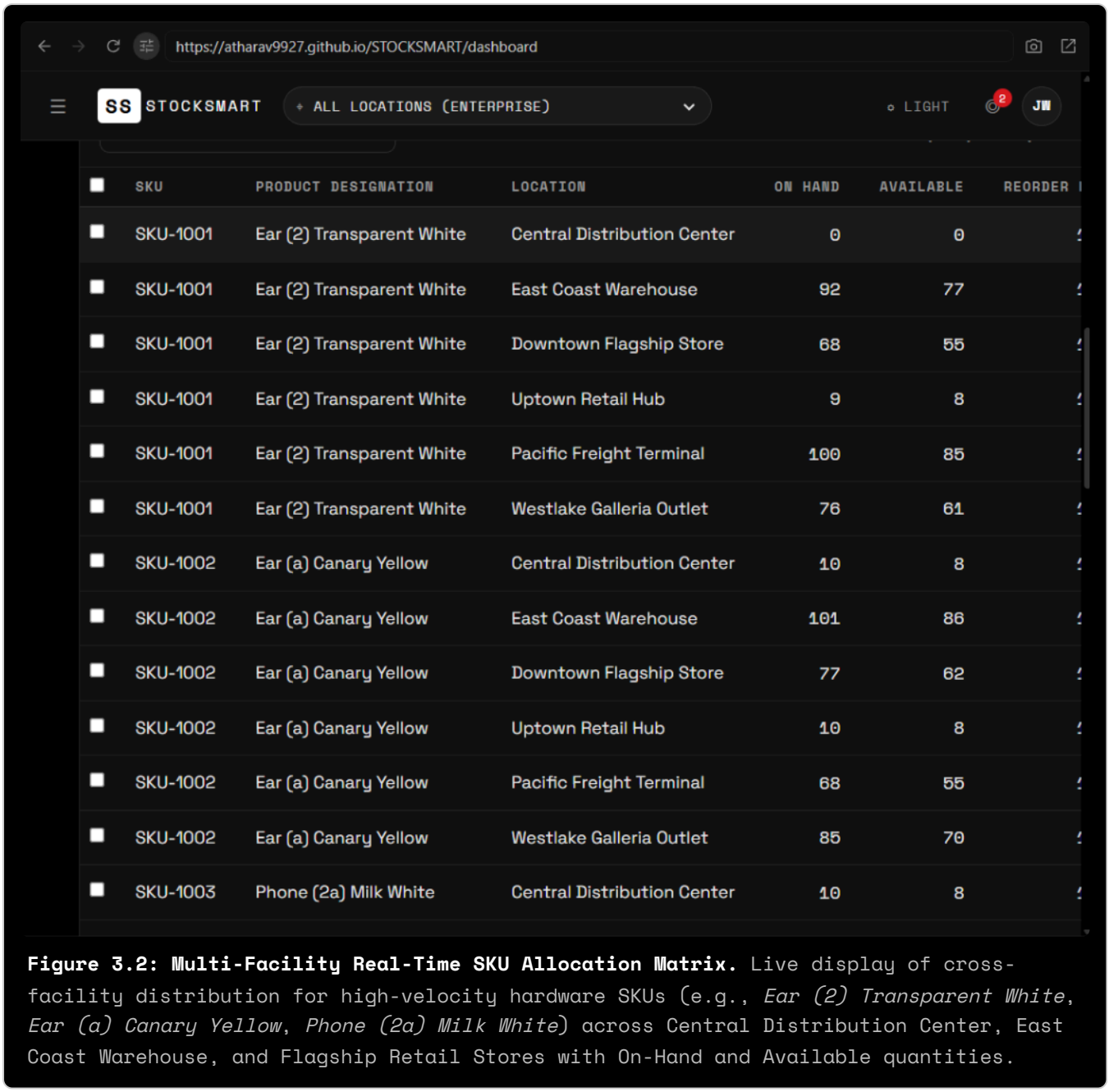


Figure 3.1: Operations Command Center & Telemetry KPIs. Live URL: `https://atharav9927.github.io/STOCKSMART/dashboard`. Features Nothing OS dot-matrix numerals (`Doto` font) displaying Total Inventory Value ($22.6M with positive green sparkline), Low Stock Items (240 SKUs with warning red curve), Critical Stockouts (120 SKUs), and Active Purchase Orders.



SKU	PRODUCT DESIGNATION	LOCATION	ON HAND	AVAILABLE	REORDER
SKU-1001	Ear (2) Transparent White	Central Distribution Center	0	0	
SKU-1001	Ear (2) Transparent White	East Coast Warehouse	92	77	
SKU-1001	Ear (2) Transparent White	Downtown Flagship Store	68	55	
SKU-1001	Ear (2) Transparent White	Uptown Retail Hub	9	8	
SKU-1001	Ear (2) Transparent White	Pacific Freight Terminal	100	85	
SKU-1001	Ear (2) Transparent White	Westlake Galleria Outlet	76	61	
SKU-1002	Ear (a) Canary Yellow	Central Distribution Center	10	8	
SKU-1002	Ear (a) Canary Yellow	East Coast Warehouse	101	86	
SKU-1002	Ear (a) Canary Yellow	Downtown Flagship Store	77	62	
SKU-1002	Ear (a) Canary Yellow	Uptown Retail Hub	10	8	
SKU-1002	Ear (a) Canary Yellow	Pacific Freight Terminal	68	55	
SKU-1002	Ear (a) Canary Yellow	Westlake Galleria Outlet	85	70	
SKU-1003	Phone (2a) Milk White	Central Distribution Center	10	8	

Figure 3.2: Multi-Facility Real-Time SKU Allocation Matrix. Live display of cross-facility distribution for high-velocity hardware SKUs (e.g., *Ear (2) Transparent White*, *Ear (a) Canary Yellow*, *Phone (2a) Milk White*) across Central Distribution Center, East Coast Warehouse, and Flagship Retail Stores with On-Hand and Available quantities.

4. Core Operational Modules (42 Routes)

MODULE DOMAIN	ROUTE PREFIX	KEY CAPABILITIES & WORKFLOWS
Public Access	`/`, `/login`	Nothing OS landing page with live telemetry, operator login form, credential recovery, 1-click role bypass.
Command & Telemetry	`/dashboard`	Executive enterprise overview, multi-location facility switcher, financial valuation sparklines, action quick-links.
Catalog Master	`/catalog/*`	520 SKU directory, drag-and-drop category taxonomy CDK tree, thermal Avery/Zebra barcode label printer sheets.
Stock Matrix & Transfer	`/inventory/*`	Cross-facility 2D stock matrix, inter-warehouse replenishment transfers, shrinkage adjustments, blind cycle count audits.
Fulfilment & POS	`/orders/*`	3-step barcode verified pick-pack-ship outbound queue, reverse logistics RMA returns, instant cashier POS terminal.
Procurement & RTV	`/procurement/*`	Multi-line purchase order generation, goods receiving dock short/damaged delivery tallies, vendor return (RTV) claims.
Auditing & Shrinkage	`/reports/*`	Inventory turnover ratios, stockout risk forecasting, ABC velocity distribution, damage/pilferage shrinkage ledger.
Administration & RBAC	`/admin/*`	Operator management, 5-level role permission matrix, facility settings, immutable ledger backups.

5. Security Model & Role Clearance Matrix

StockSmart implements a strict role-based access control (RBAC) hierarchy. Evaluators can access the live system using the pre-seeded operator profiles (Standardized Password: `password`):

ROLE CLEARANCE	SYSTEM EMAIL	OPERATIONAL SCOPE
System Admin	`admin@stocksmart.io`	Unrestricted global access: RBAC matrix configuration, data backups, facility setup, immutable audit ledger.
Inventory Manager	`manager@stocksmart.io`	Stock adjustments, transfer approvals, discrepancy write-offs, cycle count reconciliations, shrinkage audits.
Procurement Buyer	`buyer@stocksmart.io`	Supplier vendor scorecards, PO requisitions, replenishment thresholds, vendor return (RTV) credit note claims.

ROLE CLEARANCE	SYSTEM EMAIL	OPERATIONAL SCOPE
Warehouse Clerk	`clerk@stocksmart.io`	Scoped facility clearance: Camera & wedge barcode lookup, pick-pack-ship scanning, blind count entry.
Auditor / Viewer	`viewer@stocksmart.io`	Read-only clearance: Financial inventory valuation, stockout forecasts, stock allocation matrix, turnover metrics.

6. Conclusion

StockSmart successfully demonstrates an industrial-grade solution uniting aesthetic restraint with full-stack capability. By leveraging Angular 17 Standalone Signals, Spring Boot 3.3.5, and dual deployment targets (GitHub Pages & Render), the platform provides sub-50ms operational response times, robust auditability, and zero barrier-to-entry for evaluators.